

--- title: Which local model writes the best Finnish? A blind test settles it project: portfolio date: 2026-07-02 type: research ---

Which local model writes the best Finnish? A blind test settles it

This is the third measurement round in this RAG's Finnish saga, and the one that finally measured the right thing. Three 8B local models (`qwen3:8b` , `llama3.1:8b` , `Poro-2-8B`), 30 Finnish questions, 540 generations, three measurement layers, zero cost, no LLM judge anywhere. The capstone was a blind ranking: 30 rounds where a native speaker graded pure linguistic naturalness without knowing which model wrote what. Poro, twice dismissed by earlier metrics, won 26 of 30 rounds. Friedman $p < 0.0001$.

Why a third round

The earlier experiment (the Finnish experiment post) concluded that Poro wins nothing on synthesis. But it left one loose thread, in its own words: `"Its only edge is qualitative: more natural inflected morphology ("Astro 6:sta", "TypeScriptistä"): an edge for a human reader, not for the metric."` A checklist metric is blind to the one thing a Finnish-built model exists for. This round built an instrument for that human reader.

The question, precisely: which model produces the best Finnish `_language_`. Not the most correct answers. Correctness is still measured, but only as a disqualification floor, so that a hallucinating model cannot win on style.

Method: three layers, no LLM judge

An English-centric judge model grading Finnish is the exact bias under study, which disqualified LLM-as-judge from the start. The three layers:

1. **Fact floor (deterministic)**. 30 must-retrieve Finnish questions over this portfolio's corpus, each with verbatim-verified checklist facts. 119 in total, every one confirmed present in the corpus before any model ran.
2. **Voikko linguistic analysis (deterministic)**. libvoikko spelling and morphology over every answer, with two-tier token classification (tech identifiers, proper nouns and English loans separated from genuine Finnish errors) plus per-answer language detection. The instrument was calibrated on 168 human-approved Finnish UI strings from this site's own translations. Voikko flags 7.2% of that known-good Finnish, so a model's error rate means something only above that floor.
3. **Blind human ranking (the gold layer)**. 30 rounds: 10 questions, 3 independent generations each. The three models' answers were anonymized and shuffled with a seeded RNG, and the key stayed sealed until all 30 rankings were recorded. The judge (me, the native target reader of this deployment) ranked naturalness and fluency only. Ties were allowed.

The harness discipline carried over from the earlier rounds: a single variable (`LLM_MODEL`), lock-asserted `top_k=6`, temperature 0.4 and `num_ctx` 8192, identical retrieval for every arm, a per-arm instrument fingerprint, and the rate-limiter loopback exemption that the previous round's correction forced. Three runs per model per prompt version, with the per-run spread reported rather than hidden.

Result I: the language router tested itself first

The first thing the new eval set exposed was not model quality. Asked in Finnish with an English retrieved context, roughly half the answers came back in English, and the initial read was model drift. The capture data says otherwise: 15 of the 30 questions never reached the Finnish path at all. The pipeline's language router, an `ä/ö-and-function-word`

heuristic whose weakness on code-heavy Finnish the earlier experiment had already documented, classified them as non-Finnish and served them the full English-forcing prompt. Their English answers are obedience, not drift. (This correction was caught by the code review of this study's own PR, after the first version of this post was written. The process keeps working.)

On the 13 scored questions the router genuinely routed to Finnish, the picture changes completely:

model	one anchor	three anchors
Poro	33/39 (85%)	39/39 (100%)
llama	39/39 (100%)	39/39 (100%)
qwen3	34/39 (87%)	36/39 (92%, two empty answers)

The prompt fix stands: the Finnish path had one language anchor where English had three, and mirroring them, with an explicit instruction to translate the explanation while keeping product names and code identifiers as they are, took Poro from 85% to 100%. But the honest denominator changes the conclusion. With a correct router and a symmetric prompt, Poro does not drift at all. And llama's apparent adherence lead in the overall numbers came partly from disobeying the English-forcing instruction on misrouted questions. Obedience and good Finnish are different virtues.

Two of the 30 questions never reached any model at all: the English embedder retrieved them below the relevance gate. That is the same Finnish-retrieval limit the first experiment measured, still biting. Since the fixed fallback reply is identical for every model, those two were excluded from ranking aggregates, leaving 28 questions and 84 scored answers per model per version.

Result II: when they write Finnish, the spelling is a tie

Voikko error rate on genuinely-Finnish answers, strengthened prompt: Poro 3.3%, qwen3 4.0%, llama 5.8%. All three sit below the 7.2% human-baseline floor. In other words, when any of these models writes Finnish, its spelling and morphology are indistinguishable from human-approved Finnish at this sample size. The differences live where a spell-checker cannot see: rhythm, word choice, idiom.

The instrument itself needed honesty first. A naive Voikko pass scored all three models at roughly 60% "errors", and the human-approved calibration strings at 18%, because it counted tech-Finnish like "TypeScriptistä" and "RAG-järjestelmän" as typos. Two-tier token classification fixed the ruler before it read anything.

Result III: the blind test

30 rounds, judged blind, naturalness only:

model	mean rank	firsts /30	sole firsts	sole lasts
Poro-2-8B	1.37	26	17	0
qwen3:8b	2.23	9	3	9

model	mean rank	firsts /30	sole firsts	sole lasts
llama3.1:8b	2.40	6	1	11

Friedman $\chi^2 = 22.85$ (df 2), $p < 0.0001$. Kendall's $W = 0.38$. Pairwise sign tests: Poro beats qwen3 20 to 3 ($p = 0.0005$) and llama 22 to 1 ($p < 0.0001$), while qwen3 against llama is a coin flip (13 to 10, $p = 0.68$). Poro was best or tied-best on 9 of 10 questions and was never the sole worst answer in any round.

The earlier metric had scored qwen3 and Poro identical, both at roughly 93% "substantive grounded Finnish". The blind test says a native reader separates them 20 to 3. Both measurements are correct. They measure different things, and only one of them is the thing a reader experiences.

Result IV: the fact floor, and one instructive hallucination

Checklist-fact coverage came out around 52% for all three, with differences inside the noise. No model won on style while losing on substance, so the floor held.

One question earned a case study. Asked in Finnish where Mikko got his first paid programming job, all three models confidently answered "Vuohiliitto", which is one of his own projects, not an employer. The correct answer (Kasvu Labs Oy) sits in the corpus's CV. Retrieval simply never surfaced that chunk for the Finnish query; the retrieval log shows a miss on every arm. The models improvised from the wrong chunks. The hallucination's root cause was retrieval, not generation. Grounding is a chain, and the embedder is still its weakest Finnish link.

Findings

1. **"Poro is worst" measured the wrong thing.** Task checklists and containment probes rank models on dimensions where Finnish-specialisation is invisible. Graded blind on the language itself, Poro is decisively best.
2. **The instrument drifted more than the models.** Half of the apparent language drift was my own router forcing English on code-heavy Finnish questions; most of the rest was the under-anchored prompt, and the triple anchor eliminated it for Poro entirely (85% to 100% on the honestly-routed subset). What remains is qwen3's residual 92%.
3. **A Finnish RAG needs three repairs, not one:** a language-anchored prompt (done), a language router that survives code-heavy Finnish (open), and Finnish-capable retrieval (open). The English embedder both drops questions at the gate and starves grounding, as the Kasvu Labs case shows.
4. **The deployment decision:** Poro plus drift mitigation for the Finnish path, and the current default model for English. Pick the best Finnish writer and patch its failure mode with architecture, rather than picking a weaker writer for its obedience.

Limitations, recorded

- One judge. The native target reader of this specific deployment is the right
- judge for this decision, and still a sample of one.
- 30 blocks come from 10 questions with 3 generations each, so same-question
- generations are not fully independent. The pairwise margins of 20 to 3 and

- 22 to 1 survive this caveat comfortably.
- The blind set is conditional: only questions where all three models stayed in
- Finnish could be ranked. Poro's quality win presumes its drift is managed.
- One quantization (Q4_K_M), one GPU, this corpus and this prompt. The result
- is a deployment decision, not a leaderboard.

The series

1. The harness post: the measurement discipline.
2. The Finnish experiment: capability, plus the published containment error and its correction.
3. The methodology lesson: how the error was caught, with unrecorded variables and a rate-limiter contaminant.
4. This post: the blind test that finally measured the thing a reader experiences.